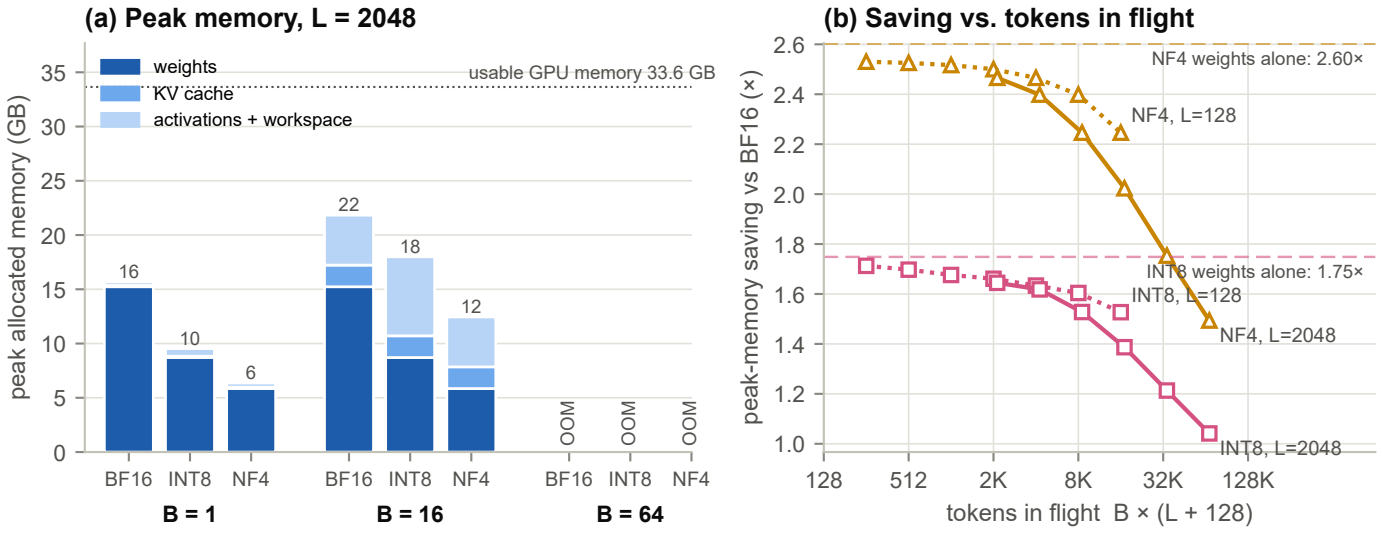
(a) Peak memory, L = 2048
weights
KV cache
activations + workspace
usable GPU memory 33.6 GB
peak allocated memory (GB)
0
5
10
15
20
25
30
35
16
10
6
22
18
12
OOM
OOM
OOM
BF16
INT8
NF4
BF16
INT8
NF4
BF16
INT8
NF4
B = 1
B = 16
B = 64
(b) Saving vs. tokens in flight
peak-memory saving vs BF16 (×)
1.0
1.2
1.4
1.6
1.8
2.0
2.2
2.4
2.6
NF4 weights alone: 2.60×
NF4, L=128
INT8 weights alone: 1.75×
INT8, L=128
NF4, L=2048
INT8, L=2048
128
512
2K
8K
32K
128K
tokens in flight B × (L + 128)